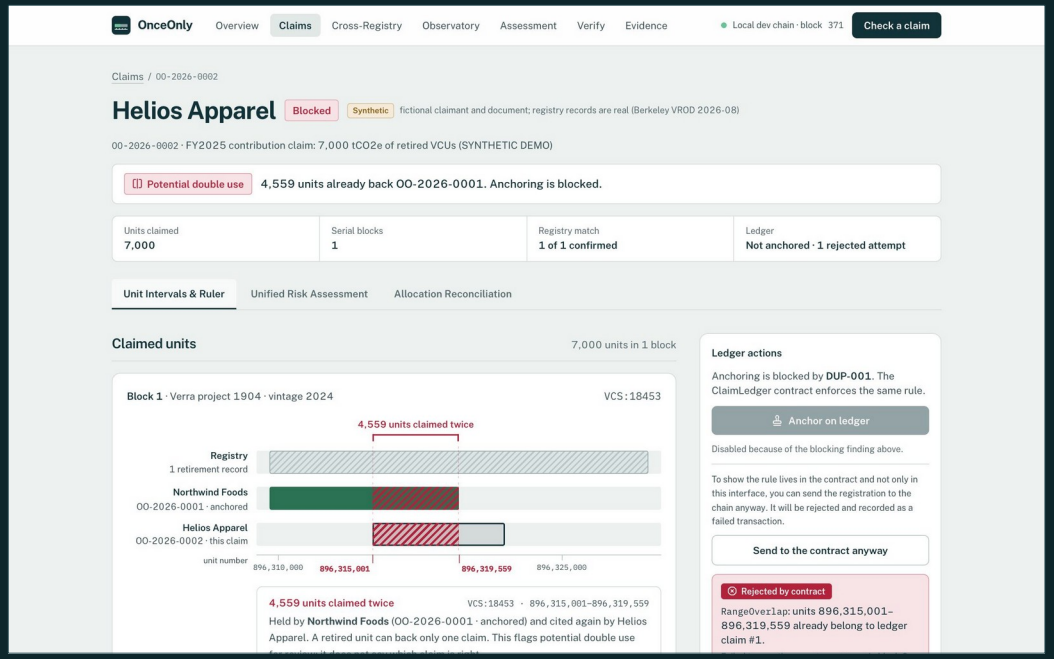
DATANOVA 2026 GLOBAL DATA-THON

# OnceOnly

## Every retired tonne, claimed once.

A carbon-credit claim integrity and investigation platform. It reconciles public climate claims with real registry retirements, explains every signal with its evidence and limits, and anchors claims on a contract that refuses to register the same unit twice.

**Siddhant Patil**
`github.com/Sidd927/OnceOnly-DataNova`




Real Verra retirement (Berkeley VROD 2026-08, batch 18453). Synthetic claimants. Local Hardhat development chain.

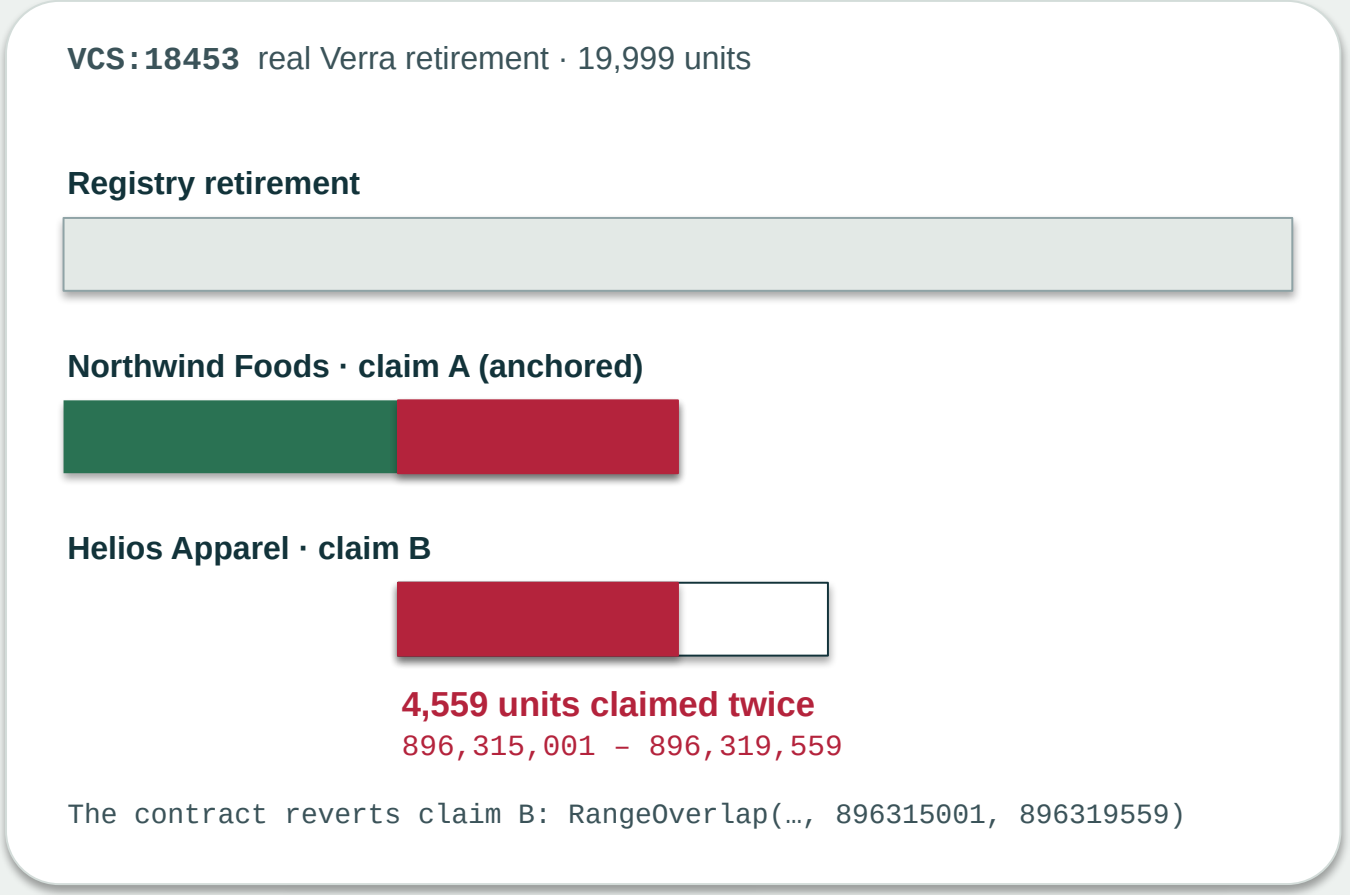
THE PROBLEM

# A retirement happens once. Claims about it happen everywhere.

**Registries stop a credit being retired twice.** Nothing checks whether the same retired units then back two companies' public climate claims.

**Brokers split one retirement across many buyers,** and claims cite it in reports, labels and websites, each one valid when read alone.

**Hashes and string matching miss it:** overlapping claims usually cite different serial strings in different documents.



WHY SIMPLE CHECKS FAIL

# The evidence is fragmented, and the obvious shortcuts break

## 0.333

**recall of exact serial / document-hash matching**

It misses every reformatted, partial, contained or superset overlap in the 270-pair evaluation.

## 159,511

**spurious overlaps if the batch is ignored**

Unit numbers restart in every issuance batch (11,631 Verra + 147,880 Gold Standard adjacent pairs). Ranges must be keyed by registry and batch.

## 62.7 %

**of analysed retirements name no beneficiary**

Undisclosed is a disclosure fact, not misconduct, but it limits who can be linked to which units.

## 2 roles

**broker vs principal, parent vs subsidiary**

"Retired on behalf of" and related companies look like conflicts to a naive interval check, and similar names are not legal identity.

Sources: eval/results/latest.json and registry_audit.json (full Berkeley VROD 2026-08: 493,518 rows); beneficiary share from the 37,335-retirement demo subset.

OUR SOLUTION

# From a claim document to an explained, verifiable result

01
### Read
Deterministic serial parser with provenance on every block; SHA-256 of each document

02
### Reconcile
Serials become unit intervals in (registry, batch) namespaces, matched to real retirements

03
### Allocate
Interval geometry + entity resolution: split, broker, related parties or conflict

04
### Profile
Disclosure, concentration, fragmentation, timing and cross-registry context

05
### Assess
16 deterministic rules, 6 scopes; evidence, counterevidence, missing data, next step

06
### Anchor, verify
ClaimLedger refuses overlapping units; per-check public verification

WHO IT IS FOR

**Sustainability auditors**
check a client's claim against registry records and other claims

**Registry / standards reviewers**
see how retirements are split, disclosed and timed

**Carbon-market analysts**
triage projects, batches and entities with explainable signals

Highlighted steps (Allocate, Assess) are the DATANOVA additions to the original claim-ledger core.

ARCHITECTURE

# Explainable analytics off-chain; one invariant on-chain

### Data

· Berkeley VROD 2026-08 (real): 493,518 rows; demo subset 38,004 rows, 22 projects
· Non-destructive normalization: raw values kept
· Synthetic claim documents
· Curated benchmarks (124 + 36 cases)

### FastAPI backend · Python 3.12

· Claim pipeline: serials → registry rules → findings → report hash
· 2A entity resolution: normalization, intermediaries, blocking, matching
· 2B allocation reconciliation: interval algebra, 8 allocation classes
· 2C concentration · fragmentation · temporal · cross-registry
· Stage 4 assessment: 16 rules, dependency groups
· Verification: on-chain vs off-chain checks reported separately
· SQLite (claims, read-only registry DB) · content-addressed evidence

### ClaimLedger.sol

· Ranges keyed by keccak(registry:batch)
· O(1) hinted sorted list; RangeOverlap revert
· Local Hardhat chain 31337

### React 19 UI

· Overview · Claims · Cross-Registry
· Observatory · Assessment
· Verify · Evidence
· Bespoke SVG charts, no chart library

Deployment: one Docker image (private chain, contract, seeded demo, API serving the UI), about 280 MB of memory; Render blueprint prepared.

TECHNICAL DIFFERENTIATION

# Intervals, not hashes. Classes, not one red flag.

### Allocation-aware reconciliation

Six geometric and eight allocation classes separate a legitimate split, a broker acting for a disclosed principal, related companies and an unresolved beneficiary from a conflict between unrelated claimants.

### Namespace isolation

Every comparison is inside one (registry, batch). Identical numbers or project IDs across registries are an identifier collision, never the same asset.

### Defensible statistics

HHI and entropy over disclosed volume only; pooled vs within-batch Gini reported separately; leave-one-out MAD bursts; absolute vs relative small allocations.

### Explainability by construction

16 versioned deterministic rules; each finding carries thresholds, counterevidence, missing data and a next step; correlated signals grouped so they cannot double-count.

### Verification you can redo

Document, commitment, ranges, network and deployed bytecode checked separately; database lookups never counted as on-chain proof.

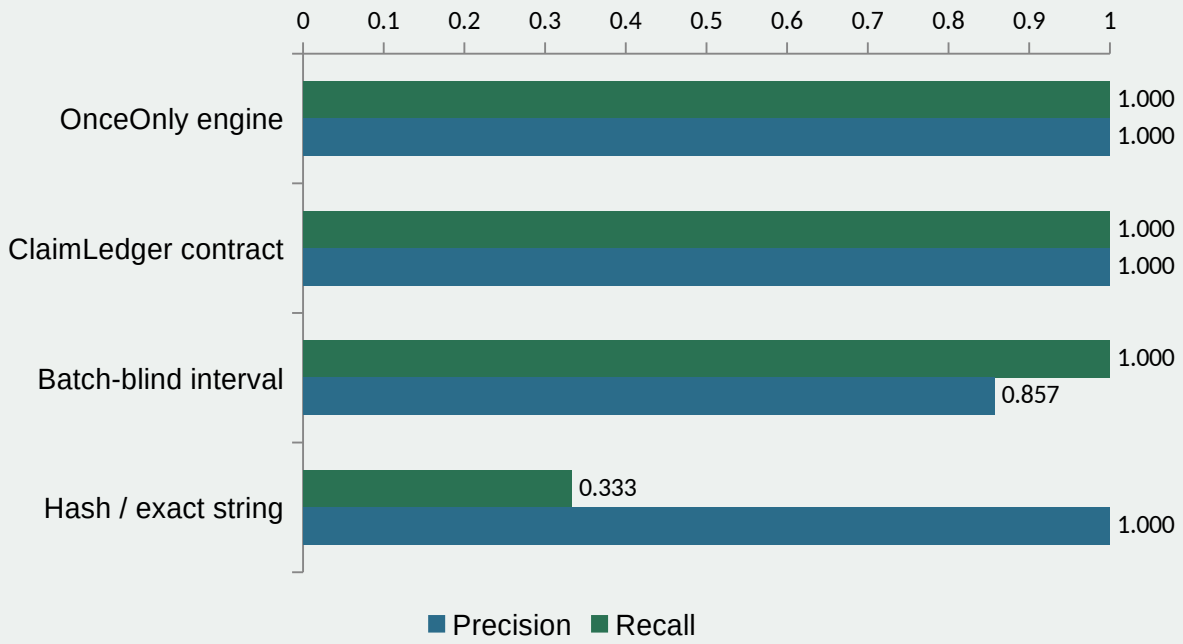
### A contract that scales

Registration gas flat at 301,452 from 10 to 200 active ranges per batch (linear v0: 832,608); a bad hint reverts, it never skips the check.

EVIDENCE AND VALIDATION

# Measured, and labelled for what each test measures

## Double use on 270 synthetic claim pairs citing real Verra ranges




Ground truth uses the engine's interval definition, so 1.000 verifies the implementation; the finding is what the simpler methods miss. Not real-world accuracy.

## Data and analysis checks

| | |
|---|---|
| Registry serials → unit ranges (full VROD) | **493,518 / 493,518** |
| Overlaps inside one batch (Verra) | **0 of 307,416** |
| Entity resolution, 124 curated pairs | **P 1.000 · R 0.964** |
| Allocation class, 36 curated scenarios | **36 / 36** |
| Serial extraction, 60 synthetic snippets | **100 % in-sample** |
| Population metrics, independent SQL re-run | **exact match** |

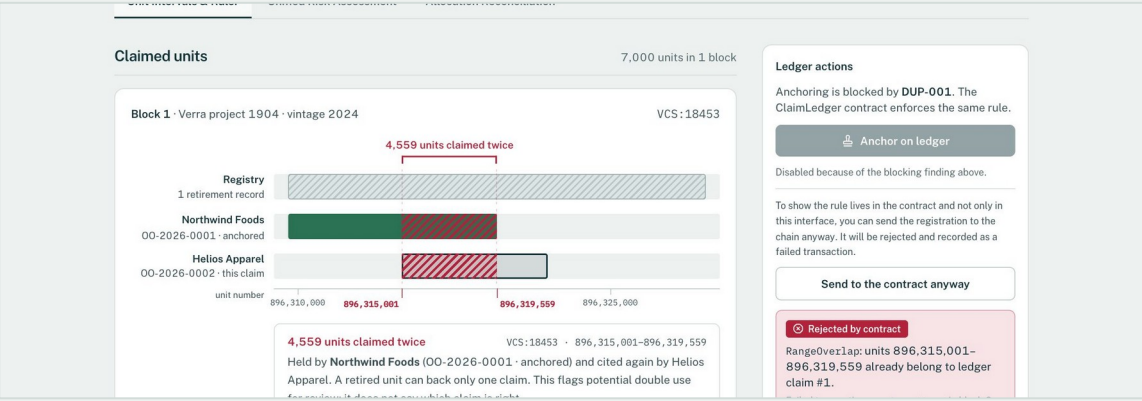
## Software tests (this release)

**271** backend · **24** contract · **45** browser · **13/13** end-to-end

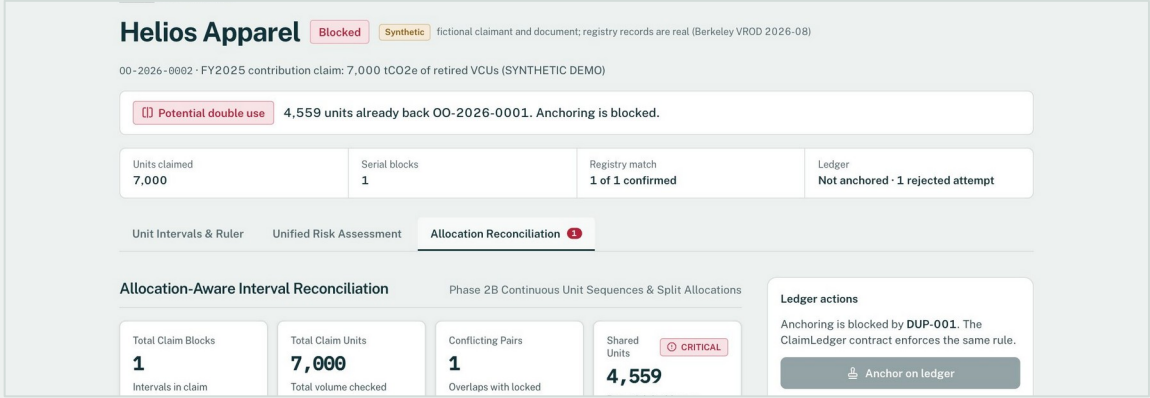
`Demo state fingerprint reproduced: c544f589…6483b`

PRODUCT DEMONSTRATION

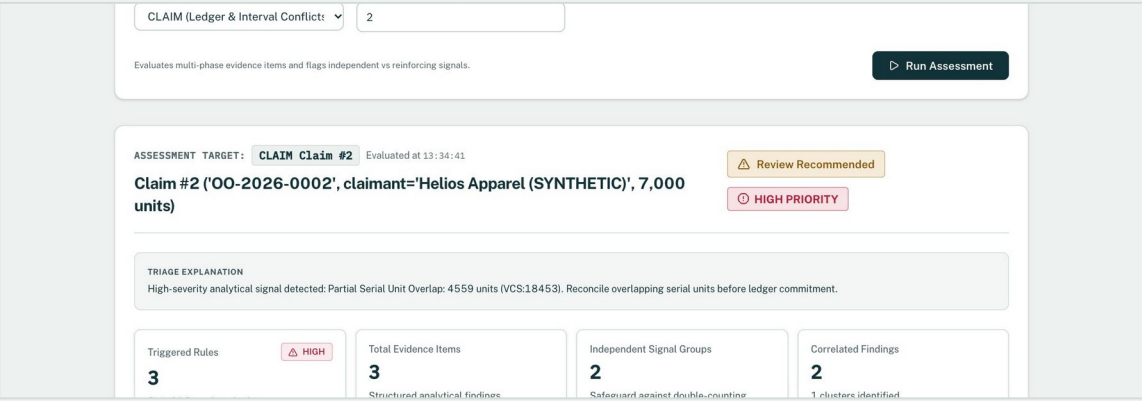
# One real retirement, two synthetic claims, every step explained



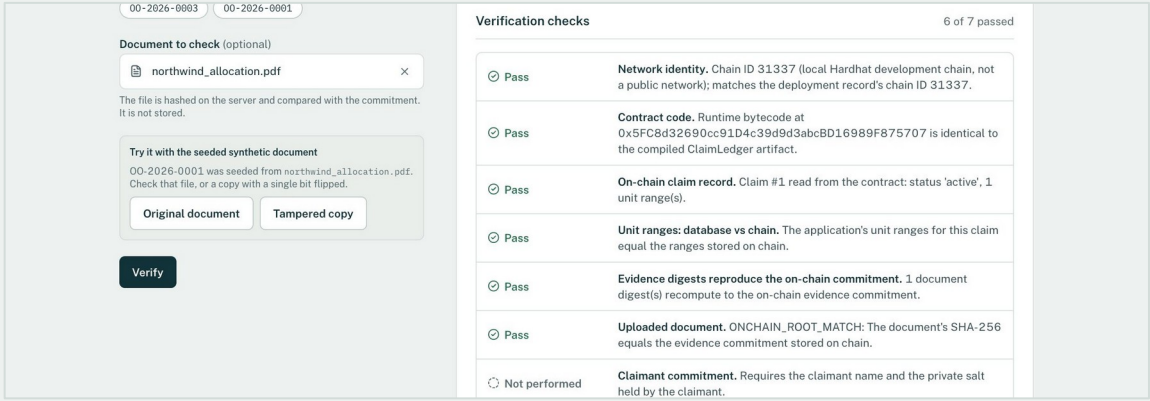

1 · The Ruler: 4,559 units already back Northwind's anchored claim; the contract rejects the second registration




2 · Allocation reconciliation: ALLOCATION_CONFLICT between different entities, with a next step




3 · Assessment: RULE-REC-002 with metrics, limitations and grouped concentration signals




4 · Verification: seven separate checks, including deployed bytecode and the document commitment

# What it verifies, what it flags, what it cannot conclude

## VERIFIES

· Serial strings parse into exact unit ranges and match registry retirement records (snapshot 2026-08-31)

· Two claims share specific units in the same registry batch

· A document is byte-identical to what was committed on chain

· The deployed contract code matches the compiled artifact

## FLAGS FOR REVIEW

· Potential double use, classified by allocation type

· Concentration, fragmentation, timing anomalies

· Candidate cross-registry project links

· Missing beneficiary information (reported, never imputed)

## CANNOT CONCLUDE

· Fraud, intent, or which claim is legitimate

· Legal identity from similar names or parent/subsidiary links

· Credit quality, additionality or permanence

· Anything about the whole market: analytics cover a 22-project subset

**Trust boundaries:** local Hardhat development chain (31337), not a public network · one trusted registrar key, no claimant signatures · synthetic claimants · static registry snapshot · AI extraction implemented but not measured · benchmarks synthetic or curated, not real-world accuracy.

IMPACT AND NEXT STEPS

# From a working prototype to a usable integrity check

## Practical value

**Catches what document review and hashing miss:** partial and reformatted overlaps of the same retired units.
**Makes analysis reviewable:** every signal shows its rule, threshold, counterevidence and next step, so an auditor can defend or reject it.
**Reproducible:** public data, open code, one-command container, deterministic demo state.
**Neutral ordering:** a contract invariant that the operator's own UI cannot bypass.

## Credible path forward

1. Public testnet deployment with claimant-signed registrations (EIP-712)
2. Per-document Merkle commitments and proofs
3. Registry data beyond the 2026-08 snapshot; more registries
4. Measure AI extraction (accuracy, cost, latency) on real documents
5. Pilot with real published claims, with consent, to measure prevalence

**Run it:** `make docker-build && make docker-run → http://127.0.0.1:7860`
**Code, data notes and methodology:** `github.com/Sidd927/OnceOnly-DataNova` · Registry data: Berkeley Carbon Trading Project, VROD v2026-08 (CC BY 4.0)